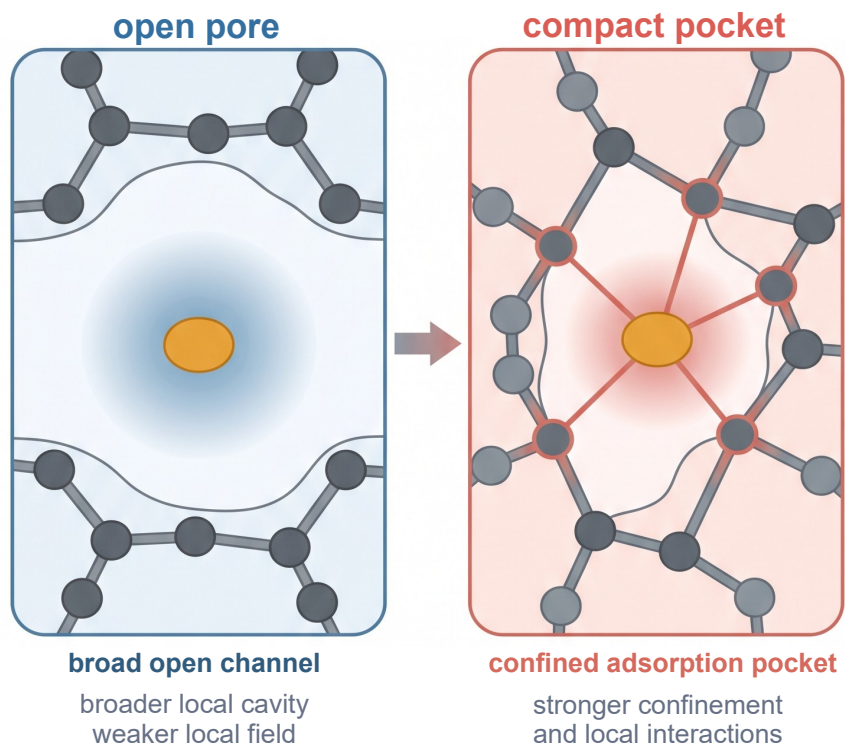


A Global pore descriptors can miss local adsorption environments



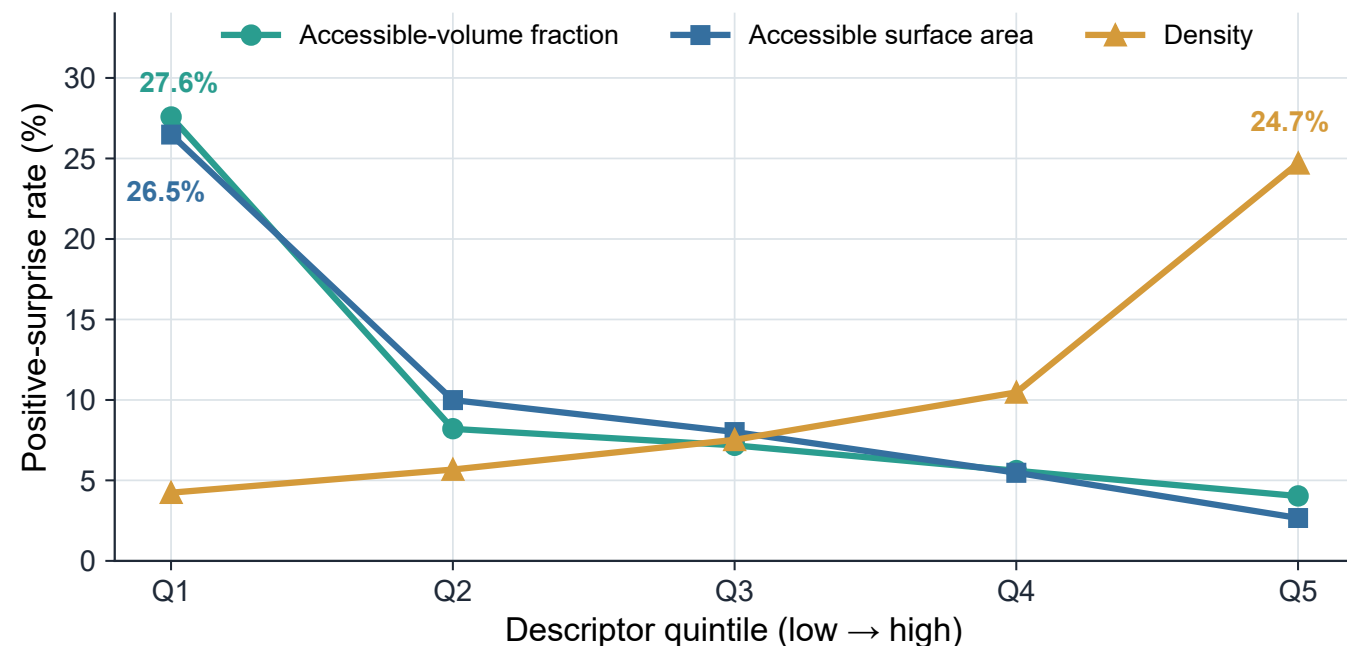
Same broad porosity signature, but different local adsorption neighborhoods.

similar global descriptors can hide different local physics

Working hypothesis

Large positive residuals may flag local adsorption physics not fully represented by global pore descriptors; they do not by themselves prove a unique microscopic mechanism.

B A monotonic chemical-regime signal across pore descriptors



C Exceptional underprediction is enriched in compact regimes

